

**Root Duality: Unifying Indexical Identity and Computational Anchoring —
A Focus on the Mechanisms of Complex Word Formation**

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Abstract:

This paper analyzes the study of root properties in constructional morphology, examining the multilingual process of complex word construction. Building on Belder & Craenenbroeck (2015), which characterizes the essence of roots as the “empty position” created by initial Merge-driven displacement, the paper presents a systematic analysis and argues that roots exhibit a dual nature as both indexical units and computational anchors.

Key words: root, empty position, index, Computational Anchoring

1. Research background and questions

Root-based research within the framework of Neo-Constructionism can be broadly categorized into three major perspectives: the Indexical View, the Ontological View, and the Structural View. The following briefly outlines the current divergences among these approaches.

(1) The Indexical View. Scholars such as Harley (2014) argue that roots function as indices. Within the syntactic derivation, Harley treats the root (ROOT) as an arbitrary numerical index that distinguishes each root from all others. She posits that a root may take on different phonological realizations in distinct syntactic environments. Since the semantic contribution of a root is highly variable—perhaps analogous to a phenomenon of “semantic substitution”—roots cannot be identified solely by their phonological or semantic content. Accordingly, Harley proposes that roots should be treated as purely structural objects in syntax,

carrying neither phonological nor semantic features, but serving as computational building blocks.

(2) The Ontological View. Scholars such as Marantz (1997) and Embick (2004) maintain that roots are endowed with inherent ontological properties. Marantz, for instance, classifies roots into entity roots (denoting objects), state roots (denoting properties), and event roots (denoting actions). Hovav and Levin (2010) further distinguish between manner and result roots, arguing that a verb's syntactic behavior is determined by the ontological type of its root.

(3) The Structural View. Proponents including Acedo-Matellán and Mateu (2015) as well as Borer (2013) contend that the semantic interpretation of roots is not inherent but derived from syntactic structure. In this view, roots are category-free abstract units whose interpretation wholly depends on their syntactic configuration and their attachment to functional morphemes such as *v* or *n*. Belder and Craenenbroeck (2015), building on principled assumptions about Merge, identify four defining properties of roots: (a) roots have no grammatical features, (b) roots have no syntactic category, (c) roots are defined structurally rather than lexically, and (d) roots are dominated by functional material (rather than the other way around). (Belder and Craenenbroeck 2015)

We will briefly analyze the three views mentioned above. Regarding the Index Theory, Marantz (2024) points out that in syntax, indices are not elements accessible to syntactic operations—the syntax does not recognize things like #53 or #67. The root should be treated as a kind of baseline, such that the morphological alternation from “go” to “went” begins from the index. Therefore, we may assume that the underlying nature of roots is phonological form. This view, in fact, provides the ontological foundation of the theory itself. As for the Structural View, Borer (2013), within the Exo-Skeletal Model, argues that roots are devoid of any intrinsic meaning; root semantics arises at a certain stage of syntactic derivation, where encyclopedic information is accessed. This view aligns with Belder & Craenenbroeck (2015), who also identify four fundamental properties of roots.

The present study follows Belder and Craenenbroeck (2015) in adopting the

view that “We show that as a by-product of this asymmetry, the first Merge operation in each cyclic domain—called Primary Merge here—creates a radically empty structural position at the foot of the structure in which a root can be inserted at the level of Vocabulary Insertion.” While various linguistic perspectives on the nature of roots (ROOT) differ in focus, they all contribute valuable insights into our understanding of root properties. The indexical view emphasizes the function of roots as serving interface operations; the ontological view regards roots as the morphological baseline; and the non-interpretive view considers roots as the semantic baseline. Building on these distinctions, I argue that the functional contribution of roots in language generation should be understood as stage-dependent. Specifically, once syntactic computation concludes and the derivation proceeds to the Sensory-Motor (S-M) and Conceptual-Intentional (C-I) interfaces, the root simultaneously assumes the dual roles of an indexical unit and a computational base.

This paper, grounded in Belder and Craenenbroeck (2015), reexamines the inherent properties of roots and proposes a theory of their dual nature. It further explores how this duality shapes the word-formation mechanisms of compounds. The remainder of the article is structured as follows: Section 2 elaborates on the dual-nature hypothesis of roots; Section 3 discusses the impact of roots as indexical units and computational bases on the derivation of compounds; Section 4 briefly considers the implications of root duality for syntactic structure building; and Section 5 concludes the study.

2. What Is a Root?

What is a root? The author believes this to be the premise for discussing the nature of roots. Belder and Craenenbroeck (2015) point out that under the frameworks of the Minimalist Program and Distributed Morphology (DM), roots are defined through syntactic structure rather than the lexicon. They propose the view of Asymmetric Primary Merge and note that during the process of syntactic building, Primary Merge is directional, always merging a set of grammatical features (such as feature set $\{\alpha\}$) with an empty set ($\emptyset$), forming an ordered pair

$\langle \{\alpha\}, \emptyset \rangle$. The projection and labeling come from $\{\alpha\}$. The empty position created by Primary Merge is the site where the root is inserted later¹. They further consider the root as a "placeholder" in the syntactic structure, lacking inherent grammatical information and category, with categorization assigned by a categorizing morpheme. They also discuss and demonstrate that roots have four distinct properties:

(a) roots have no grammatical features, (b) roots have no syntactic category, (c) roots are defined structurally rather than lexically, and (d) roots are dominated by functional material (rather than the other way around).

(Belder and Craenenbroeck 2015)

I endorse Belder & Craenenbroeck (2015) and propose that roots (ROOT) are interface-identification indices that are also staged. In the syntactic operation phase (when the syntactic terminal inventory in List 1 governs operations), their nature should be an empty position, as Belder & Craenenbroeck (2015) describe the “We show that as a by-product of this asymmetry, the first Merge operation in each cyclic domain—called Primary Merge here—creates a radically empty structural position at the foot of the structure in which a root can be inserted at the level of Vocabulary Insertion.” radical empty structural position located at the bottom of the structure. In the morphology/spelling-out phase (Vocabulary Insertion), the root can be inserted into this position. I further argue that after synchronous merging and uniform operations, this empty position provides an interface index for late insertion, functioning as an index unit, this index unit is generated during the Spell-Out stage. This index unit finishes linearization during the morphological operation phase (where List 2 supplies the word-item inventory for morphology). At this stage, the root serves as the basis for inflectional morphophonology, i.e., the root’s phonological form provides the morpho-phonological interface basis for alternations, albeit with the phonological form being a late fill-in². Likewise, during

¹ For the delayed insertion of roots, see Harley (2014:244).

² Marantz’s idea that the morphophonological adjustment from “go” to “wen(t)” begins with the index, so the underlying form of the root has a phonological realization, with the phonology filled in at a late stage.

the encyclopedic Knowledge Base phase (where List 3 provides encyclopedic knowledge), the root also completes a semantic operation baseline. See Figure 1.

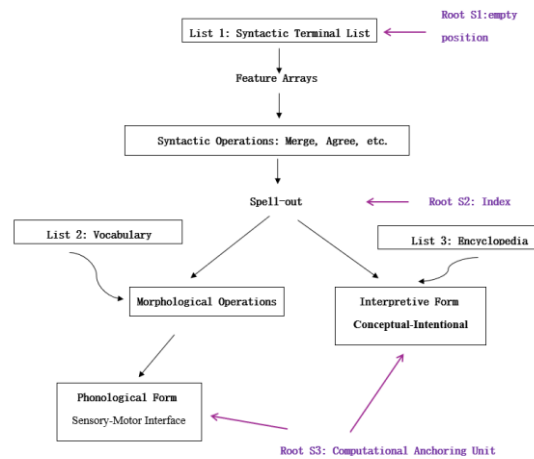


Figure 1 schematically represents this phase-based derivational process.

In sum, I regard ROOT as a staged element that is fundamentally unified as both an indexical unit and a computational anchor.

3. Root indices and Computational Anchoring (index function and baseline)

During the process of syntactic building, the stage of Primary Merge gives rise to what can be termed as “Null Derivation.” At this initial point of syntactic construction, no substantial lexical item is yet introduced; rather, the structure emerges from the merger with the “null set.” Consequently, the derivation at this earliest stage is “empty,” devoid of both phonological and semantic content. This empty position functions as an index, which is only filled with phonological and semantic values upon transfer to the interfaces following word formation. Following Marantz (2007), it is possible to treat words themselves as phases, such that, at Primary Merge, Null Derivation arises when a specific syntactic feature bundle (e.g., a feature set {a}) merges with the null set ($\emptyset$) to constitute a phase. In this configuration, the head of the phase is the syntactic feature bundle responsible for the null derivation (such as feature set {a}), while the transferred interface component is the null set. Word formation, in this view, is also bound by the Phase Impenetrability Condition (PIC).

In Building, this “empty position” (which we denote ROOTc, the initial root) exhibits two properties: operability (OY) and baseline startup capability (BSC). These properties are realized by the structure created by Merge³ and govern how roots combine with other elements to form words. After the operation concludes, the initial root position remains an interface-late phonology and meaning slot, giving rise to a baseline startup, becoming the morpho-phonological and semantic computation anchor, and is identifiable at the Spell-out stage to form indices. Given that compounding constitutes a major mechanism of word formation in Mandarin Chinese (see Chen & Liu 2020), we will illustrate these points with the compounds “da huang gou” (大黄狗, big yellow dog) and “huang da yi” (黄大衣, yellow coat).

(1) [_#√huang [_{nP}<√da √yi>,n]]

[_#√黄[_{nP}<√大 √衣>,n]]

yellow coat

(2) <[_#√da [√haung √gou]],n>

<[_#√大[√黄 √狗]],n>

big yellow dog

(1) huang + dayi (“yellow coat”) is an outer-domain compound, where the co-component for coat is directly transferred from the yellow domain. (2) da huang gou (“big yellow dog”) is an inner-domain compound, so big and yellow attach within the root-domain. The coat, under the Phase Impenetrability Condition (PIC), cannot be operated on further.

In the first phase of roots (Phase S1 as in Figure 1), the roots “huang(yellow), da (big), yi(coat)” and “da(big),huang(yellow),gou(dog)” are merged at List 1. We regard this as an initial merge between an initial root (ROOTc) and a category-less morpheme n. According to the two features operability (OY) and baseline startup capability (BSC), the following changes occur:

³ This corresponds to the claim by Belder & Craenenbroeck (2015) that the properties of roots are structurally defined, rather than through syntactic category or lexical specification.

(1a)

Primary Merge: $[\{\emptyset, \emptyset\}, n] \longrightarrow \langle \{\emptyset, \emptyset\}, n \rangle$ **【OY】**

Merge-and-sequential-labeling stage: $\{\{\emptyset, \emptyset\}, n\} \longrightarrow \langle \{\emptyset_1, \emptyset_2\}, n \rangle$

Derivation by Phase: $\{\langle \{\emptyset_1, \emptyset_2\}, n \rangle\} \quad \{\langle \{\emptyset_1, \emptyset_2\}, n \rangle\}$ **【BSC】**

At this stage, transfer to the interface the word-formation segments $\{\emptyset_1, \emptyset_2\}$ for interface operations (Phase S2 and S3 in Figure 1).

Structure Building: $[\{\emptyset\}, \langle \{\emptyset_1, \emptyset_2\}, n \rangle] \longrightarrow \{\{\emptyset\}, \langle \{\emptyset_1, \emptyset_2\}, n \rangle\}$

Merge-and-sequential-labeling stage: $\{\{\emptyset\}, \langle \{\emptyset_1, \emptyset_2\}, n \rangle\} \longrightarrow \{\{\emptyset_3\}, nP\}$

Derivation by Phase: $\{\{\emptyset_3\}, nP\}$ **【BSC】**

At this stage, the word-formation phase is derived up to its highest node, with the set $\{\emptyset_3\}$ being transferred as the morpho-syntactic phase for interface operations to complete the word-formation process.

In word-formation cases like “dayi(coat)” the initial merge yields two empty positions that merge with the productive morphemes. I argue this arises because the initial root carries operability features [+OP], whose role is to trigger the merge between the root’s empty position and other constituents. Since [+OP] is a syntactic structural feature and lacks a category, it cannot bear a label; thus, when merging with a productive morpheme, the morpheme’s category is shared and labeled accordingly. Of course, when merging with another initial root carrying [+OP], the resulting empty position (which I call the Operability Empty Set) contains two constituents (the two parts of “coat” correspond to two empty positions as in 1b)⁴. In syntactic operations, there is effectively only one empty position (the Operability Empty Set); phonological realization occurs at the interface after the split and filling. We treat the “empty position” in word-formation as the Operability Empty Set; the number of elements within this set is not tied to the count of empty positions available in the workspace, but serves as a marker of the root’s index properties. This empty structure must be constructed through merging, which also satisfies the

⁴ It should be noted that the OperNS (Operability Null Set) introduced here is indeed an empty position in the syntactic structure. We define this set precisely to reflect the index-unit function of the initial root.

principle that “In every derivational workspace, there is exactly one root, and for every root there is exactly one derivational workspace. (One Derivational Workspace, One Root(ODWOR)Belder & Craenenbroeck 2015)”.

$$(1b) \{\emptyset_1\}, \{\emptyset_2\} \longrightarrow \{\emptyset_1, \emptyset_2\} = \text{OperNS}$$

$$[\text{OperNS}, n] \longrightarrow \langle \{\emptyset_1, \emptyset_2\}, n \rangle$$

It should be noted that the subscripts 1, 2, 3 on empty positions simply reflect the order of merging, not the identification index. Accordingly, I designate this stage as the Merge-and-Sequential-Labeling stage.

In the stage $\{\emptyset_3, nP\}$, after $\emptyset_3$ merges with nP , it triggers the transfer to the interface in the syntactic construction process toward the S-M (sensory-motor) interface. During the path to the Sensory-Motor Interface, the index of the initial root and the features of the category morpheme undergo a baseline-startup activation (BSC) that prompts searching for matching vocabulary items (VIs) in List 2, i.e., in its own lexicon, for insertion. At the same time, the concept–intention system provides meaning through encyclopedic knowledge.

Word formation in “da huanggou(Big Yellow Dog)” follows the same OperNS and category-morpheme merging, with the difference that there are three initial roots. First, two of them merge to form an OperNS1, then OperNS1 merges with the third initial root to form OperNS2, and finally OperNS2 merges with a category morpheme (as shown in 2a).

$$(2a) \{\emptyset_1\}, \{\emptyset_2\} \longrightarrow \{\emptyset_1, \emptyset_2\} = \text{OperNS}_1$$

$$\{\emptyset_3\}, \text{OperNS}_1 \longrightarrow \{\emptyset_3, \text{OperNS}_1\} = \{\emptyset_3, \{\emptyset_1, \emptyset_2\}\} = \text{OperNS}_2$$

$$[\text{OperNS}_2, n] \longrightarrow \langle \{\emptyset_3, \{\emptyset_1, \emptyset_2\}\}, n \rangle$$

In Derivation Workspace: OperNS merges with category morphemes, and each OperNS can itself be treated as a derived workspace. Thus, the empty positions $\emptyset$ presented above can be regarded as arising from merges with $[+OP]$, i.e., all $\emptyset$ correspond to $\langle [+OP], \emptyset \rangle$. This aligns with the ODWOR framework. It should also be noted that the merge order for “ $\sqrt{\text{da}}(\sqrt{\text{big}})$ ”, “ $\sqrt{\text{huang}}(\sqrt{\text{yellow}})$ ”, and “ $\sqrt{\text{gou}}(\sqrt{\text{dog}})$ ” is established at the interface through encyclopedic knowledge. In the construction, because merges are pairwise, an OperNS will appear within the

OperNS itself during the syntactic process (these remain empty positions by structural necessity and are not labeled yet). Since labeling cannot occur at this stage, the final labeling and meaning assignment are activated only when the phrase transfers to the interface, at which point baseline startup capability (BSC) enables phonology and semantics to be supplied.

Our previous analysis of compound words adopts the approach where roots can merge directly (Zhang 2007; Bauke 2014; De Belder 2015). However, some scholars have challenged this view, notably Nóbrega (2015) and Chenchen Julio Song (2020), who argue that roots are defective syntactic units and cannot independently undergo syntactic operations. To address this “defect,” Nóbrega (2015) proposes the notion of “edge features,” suggesting that roots must merge with categorizing morphemes to become syntactically visible and manipulable. In contrast, Chenchen Julio Song (2020) introduces a functional head *Cat*, under which all roots first merge before undergoing further operations.

As we have pointed out above, initial merge in the syntactic derivation is always asymmetric—specifically, the merge of an abstract morpheme with an empty set. Regardless of how many initial roots are involved, there is only one empty position (an operable empty set) at the stage of syntactic derivation. These roots are subsequently filled in at the interface after phonological splitting, and the initial roots within the operable empty set appear at the spell-out stage as interface indices. The [+OP] feature we propose above, in effect, reformulates the nature of the functional head *Cat* introduced by Chenchen Julio Song (2020). We argue that the addition of such abstract morphemes is merely an ad hoc device to resolve the supposed defect of roots, enabling them to participate in syntactic operations. However, the feature that compensates for the root’s defect is, in fact, an inherent property of the structure itself—the operability ([+OP])—and does not require the creation of an additional device to mark this structural property. According to the theory of null derivation, the initial roots in the syntactic derivation stage are simply non-visible empty positions; the merge of several empty sets remains an empty set. Thus, there is no need to introduce abstract morphemes, as proposed by

Nóbrega (2015) and Chenchen Julio Song (2020), to construct asymmetric merge and enable syntactic operations. Instead, these initial roots can simply surface as indices at the spell-out stage and be interpreted at the interface, without requiring the syntactic structure to create special functional elements to resolve their visibility at the interface.

The Miracle Creed in Chomsky (2024) states: “We can entertain the doubts: the only numbers are 1, 2, and infinity. The only operations and relations that accord with SMT are unary, binary, and unbounded. Unary operations do nothing in I-language,¹² so we can omit them. Binary relations permit the fewest options for carrying a derivation forward, a sufficient reason for keeping to $n=2$ for admissible relations and for the operations that create them, unless shown otherwise.” and “If 1-language is basically a thought-generating system, it optimally should observe the principle T:[T] All relations and structure-building operations (SBO) are thought-related, with semantic properties interpreted at Cl.”. The first part provides a theoretical basis for the Null Derivation perspective, and the second part offers a theoretical basis for our analysis of compounds and roots.

We can refine the Building stage in (1a):

(1c)

Primary Merge: $[\text{OperNS}, n] \longrightarrow \langle \text{OperNS}, n \rangle$

Derivation by Phase: $\{\langle \text{OperNS}, n \rangle\} \longrightarrow \{\langle \text{OperNS}, n \rangle\}$

Transfer the OperNS to the spell-out phase (Spell-out) for index labeling and merge-and-sequential-labeling.

Merge-and-sequential-labeling stage: $\text{OperNS} \longrightarrow \langle \{\emptyset_1, \emptyset_2\}, n \rangle$

The Spell-Out phase at this point transfers the indices $\{\emptyset_1, \emptyset_2\}$ directly to the interfaces, where phonological realization and semantic interpretation take place.

Structure Building: $[\{\emptyset\}, \langle \text{OperNS}, n \rangle] \longrightarrow \{\{\emptyset\}, \langle \text{OperNS}, n \rangle\}$

Merge-and-sequential-labeling stage: $\{\{\emptyset\}, \langle \{\emptyset_1, \emptyset_2\}, n \rangle\} \longrightarrow \{\{\emptyset_3\}, nP\}$

Derivation by Phase: $\{\{\emptyset_3\}, nP\}$

The word-formation segment derives to the top node and transfers the operable word-formation segment $\{\emptyset_3\}$ to the interface for operations, completing the word

formation.

3.1: Re-examining the root properties of Chinese compounds

Chen & Liu (2020) provide a detailed discussion of the formation of four types of Chinese compounds: coordinate compounds, attributive compounds, verb-result compounds, and verb-object compounds. They note that (1) coordinate compounds are formed by merging two initial roots first, then combining with a determiner morpheme; (2) attributive compounds involve a modifier occupying a demonstrative position, followed by merger with the category-mapped component (the initial root merged with the determiner morpheme); (3) verb-result compounds consist of the merger of two category-mapped verbs (V1 and V2), where the determiner morpheme of V2 denotes a light event; (4) verb-object compounds are built after the verb phrase is lexicalized and then combined with a determiner morpheme. Aside from verb-result and verb-object compounds being constructed through merges of words or phrases, both coordinate and attributive compounds in Chinese start from roots in their derivational process. From Chinese coordinate and attributive compounds alone, one obtains that the initial root merges only with a determiner morpheme that maps onto the functional category, which is consistent with Belder & Craenenbroeck (2015).

Our approach to attributive compounds diverges from Chen & Liu (2020) in how we treat the processing of words like “dayi(coat)” Chen & Liu (2020) propose first forming “√yi(clothing)” merging with the determiner morpheme *n*, sharing the determiner morpheme *n*, and then undergoing a secondary merge with the modifier “√da” requiring a second merger. By contrast, based on the theory of Null Derivation and the ODWOR framework, we argue that for words such as “dayi(coat)” in the derivational stage, the operable empty set merges with the categorizing morpheme, where the elements inside the operable empty set are split during spell-out and serve only as interface indices. The internal modifier relations are determined by the semantic interface. The same logic applies to coordinate compounds. Of course, one could also regard each root as undergoing null

derivation followed by a Markov-style merger⁵, which then proceeds to the next derivational operation. The operable feature [+Oper] actually arises from the structural property of the root itself; it is an inherent attribute of the Merge operation rather than a syntactic constituent. Therefore, it cannot be marked, nor can it be transferred to the interfaces.

We can treat both attributive and coordinate compounds as syntactically identifiable when their grammatical categories are determined at the syntactic node. In such cases, the attributive and coordinate relations are provided by encyclopedic knowledge, and there is no need to assign a dedicated position for a modifying constituent in the structure. This is because, during word-formation derivation, the operable empty set in the word-formation segment will inherit the determiner-morpheme features (regardless of how many elements it contains) and, after labeling, be transferred to the interface.

3.2: Root properties of compounds from a comparative perspective

We illustrate with German compounds to assess whether the proposed root-building rules remain applicable.

(4)

Dachziegel (roof tile) Eisenbahn (train/railway) Geburtstag (birthday)

The three German examples in (4) are all attributive compounds, i.e., Root+XP structures. We focus on “Geburtstag (birthday),” which is formed by merging “√Geburt” with “√Tag” and the determiner morpheme “n” and then an linking element [+connect]⁶ (see (4a)).

Primary Merge: $[\emptyset, n] \longrightarrow \langle \emptyset, n \rangle$ **【OY】**

Derivation by Phase: $\{\langle \emptyset_1, n \rangle\} \longrightarrow \{\langle \emptyset_1, n \rangle\}$ **【BSC】**

The operable word-formation segment $\{\emptyset_1\}$ is transferred to the interface for operations (Phase S2 and S3 in Figure 1).

Merge-and-sequential-labeling stage: $\langle \emptyset, n \rangle \longrightarrow \langle \emptyset_1, n \rangle$

⁵ Note that Markov-style merging is a multi-dimensional merger in which independent workspaces merge and then project onto a single dimension, aligning with the ODWOR principle.

⁶ In the German word, the “s” is a linking consonant.

Structure Building: $[\{\emptyset\}, \{\{[+connect]\}, nP\}] \longrightarrow \{\{\emptyset\}, \{\{[+connect]\}, nP\}$

Derivation by Phase:

$\{\{\emptyset\}, \{\{[+connect]\}, nP\} \longrightarrow \{\{\emptyset\}, \{\{[+connect]\}, nP\}$ **【BSC】**

Transfer the $\{[+connect]\}$ segment for interface operations.

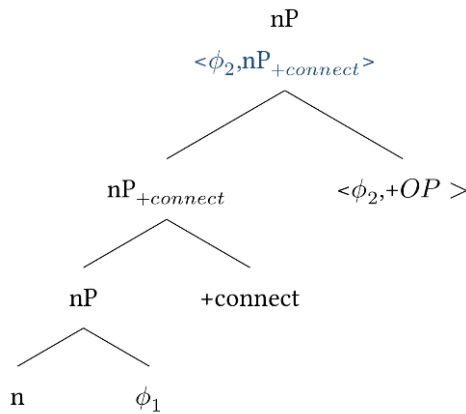
Merge-and-sequential-labeling stage:

$\{\{\emptyset\}, \{nP_{[+connect]}\} \longrightarrow \{\{\emptyset_2\}, \{nP_{[+connect]}\}$

Derivation by Phase: $\{\{\emptyset_2\}, \{nP_{[+connect]}\}$ **【BSC】**

At this stage, the word-formation segment is derived to the highest node, and the $\{\emptyset_2\}$ word-formation segment is transferred for interface operations, completing the morphological construction, as shown in (4b).

(4b)



$\emptyset_1 \longleftrightarrow /gə'bu:xt/ \longrightarrow \sqrt{\text{Tag}}$

$[+connect] \longleftrightarrow /s/ \longrightarrow s$

$\emptyset_2 \longleftrightarrow /_i ta:k/ \longrightarrow \sqrt{\text{Geburt}}$

$nP \longrightarrow \text{Geburststag}_{[+n]}$

We can summarize as follows: In the syntactic derivation stage, the root occupies an empty position, which becomes an index at the spell-out stage. At the interface, phonology is realized and interpretation is provided through encyclopedic knowledge, all starting from the root as the baseline. Therefore, ROOT is a staged unit that unifies as both an index unit and a baseline for computation.

4. Implications of Root Duality for Syntactic Construction Mechanisms in Neo-constructivism

Research into root duality has revealed that the mechanisms of word formation are, in fact, instances of phase-based derivation within syntax. This compels us to reconsider the fundamental question of what syntactic computation actually manipulates. As argued by Harley (2021), the atomic elements of syntactic construction are sets of subclass abstract features. We contend that syntactic computation directly manipulates these sets of abstract features. When the process of word formation is projected onto the syntactic domain, it involves mapping inscribable sets of abstract features (Inscriptions) into the workspace, where they are subjected to Merge (or Agree), and are subsequently transferred to the interfaces for interpretation (see Figure 2 for schematic representation).

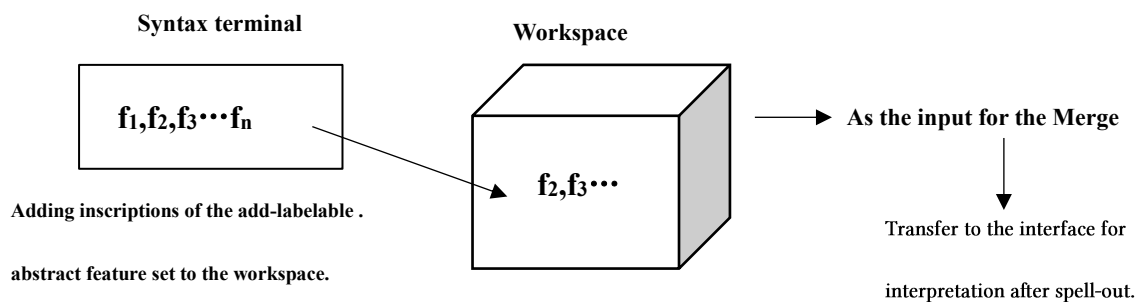


Figure 2. Language merging construction mechanism

under Neo-constructivism

In this analysis, the term “building” is intentionally favored over “generating.” The rationale is rooted in the observation that humans are not innately equipped with any particular language; instead, we possess a primordial language faculty that matures gradually and eventually enables mastery of specific languages. This gradual process encapsulates what is meant by “generation.” In contrast, when examining word and sentence formation, we are concerned with the inscription of elements onto the syntactic terminal⁷, the subsequent construction of structural

⁷ The syntactic terminal consists solely of abstract morphemes.

representations, and the incomplete valuation (“Optimality”) performed by the Vocabulary item and Encyclopaedia, with final interpretation at the interface. Thus, “building” is adopted as a term to reflect the operations involved in syntactic and lexical construction, rather than “generation.”

Furthermore, it is crucial to clarify that roots are not present on the syntactic terminal array. Rather, they arise as empty positions created by Merge; only at Spell-out are they provided with an index, subsequently serving as the computational base for construction at the interfaces.

5.Conclusion

We advocate that roots have different functions at different construction stages, and we argue that the syntactic terminal inventory in distributed morphology should not include roots. We also distinguish between generation and building, and contend that “building” is a more suitable term than “generation” when discussing word formation and sentence construction.

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